

Krishnaa Vadivel | Machine Learning Engineer | High-Performance Computing

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Summary

PhD researcher and engineer with experience building machine-learning models and deploying them on high-performance computing systems for large-scale materials simulation. Experience spans PyTorch, transformers, agentic ML, GPU acceleration, and distributed compute on national supercomputing clusters.

Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Developed equivariant graph neural networks in PyTorch and PyTorch Geometric, similar to DeepH and MACE, to accelerate optical-properties calculations in material systems with automatic differentiation.
- Built distributed and accelerator-enabled ML software in Python and C++ using CUDA, ROCm, MPI, OpenMP, and HPC environments such as Frontier and Perlmutter.
- Applied machine learning models alongside first-principles materials simulations on supercomputing clusters, connecting model development with production scientific-computing workloads.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS research in 2024, 2025, and 2026, grounding ML work in real materials-science applications.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built GPU-accelerated processing and visualization tools for large waveform datasets, providing practical experience with high-throughput data pipelines and performance-oriented engineering.
- Supported production data systems with ASP-style services and Microsoft SQL-backed tooling.

Selected Projects

Materials ML for Optical Properties: Developed equivariant graph neural networks in PyTorch Geometric to accelerate optical-properties calculations and material-energy prediction.

Matrix-Multiply AI Accelerator: Built a Verilog-based matrix-multiply accelerator project for AI-oriented datapath design and hardware-aware compute fundamentals.

Agentic Scientific RAG: Built retrieval-driven tooling that reads documentation and research papers to generate structured inputs for computational science software.

Distributed ML Compute: Combined CUDA/ROCm acceleration with HPC scheduling and scalable simulation infrastructure for ML-enabled scientific workloads.

Education

Yale University

PhD, Electrical Engineering

2021–present

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Technical Skills

ML: PyTorch, PyTorch Geometric, scikit-learn, equivariant graph neural networks, transformers, automatic differentiation

Programming: Python, C++, CUDA, JavaScript, Bash

Data / Compute: NumPy, pandas, SciPy, distributed computing, CUDA, ROCm

Systems: Linux, Docker, Docker Compose, Kubernetes, gcloud

Scientific: PETSc, SLEPc, first-principles simulation, materials modeling, physics-informed ML